

Optical flow computation in an autonomous vehicle

TX n°8116

Students: Maxime GAMONDELE, Nicolas GAUTIER

Supervisor: Jossue CARINO ESCOBAR

Context and motivation

- **Motion estimation** : both relative (to other objects) and absolute (to the environment) is among the most valuable data for autonomous vehicles and robotics

- Different sensors cover different needs: **GPS** → absolute position, **LiDAR** → relative measurements

- **Onboard cameras can provide both simultaneously**, by extracting the scene's motion field, this is optical flow

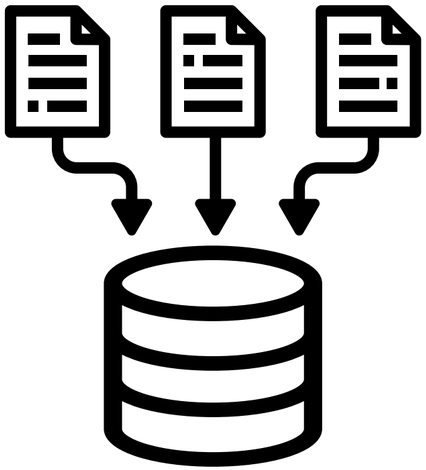
- **Project goal:** design a robust optical flow algorithm for different cameras and autonomous platforms (drones, ground robots), built and validated on custom datasets pairing camera images with motion-capture ground truth.

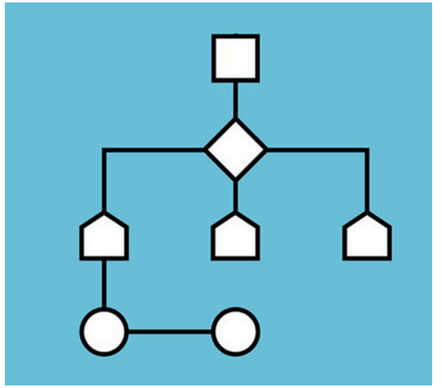






Project objectives

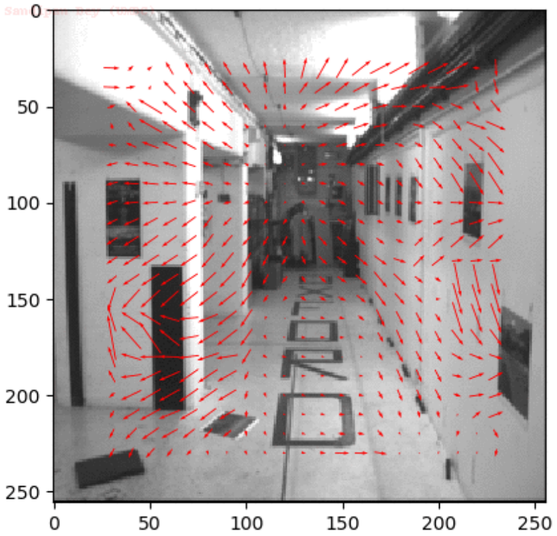




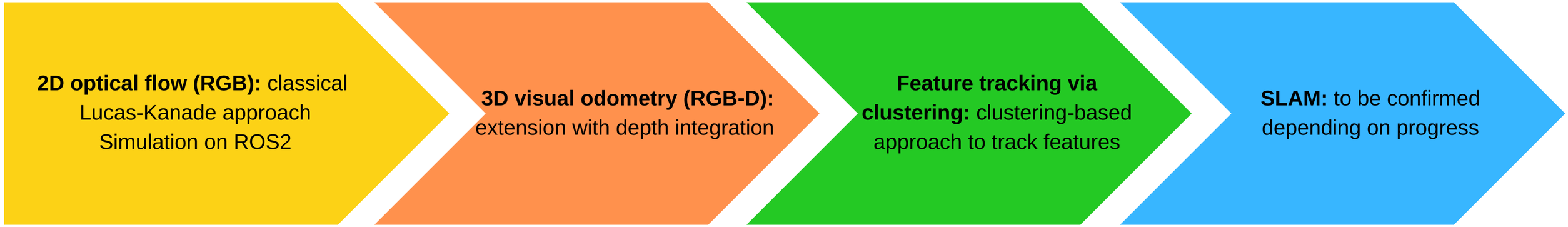
- **Develop, validate and test an optical flow algorithm** capable of running on different cameras and autonomous platforms (drones, ground robots, simulators)

- **Build datasets** combining camera images with motion-capture ground truth, used as a common benchmark

- **Compare and evaluate** several optical flow algorithms under identical conditions on Heudiasyc's platforms



Project roadmap - Current status



Where we're at right now

Project roadmap - Current status

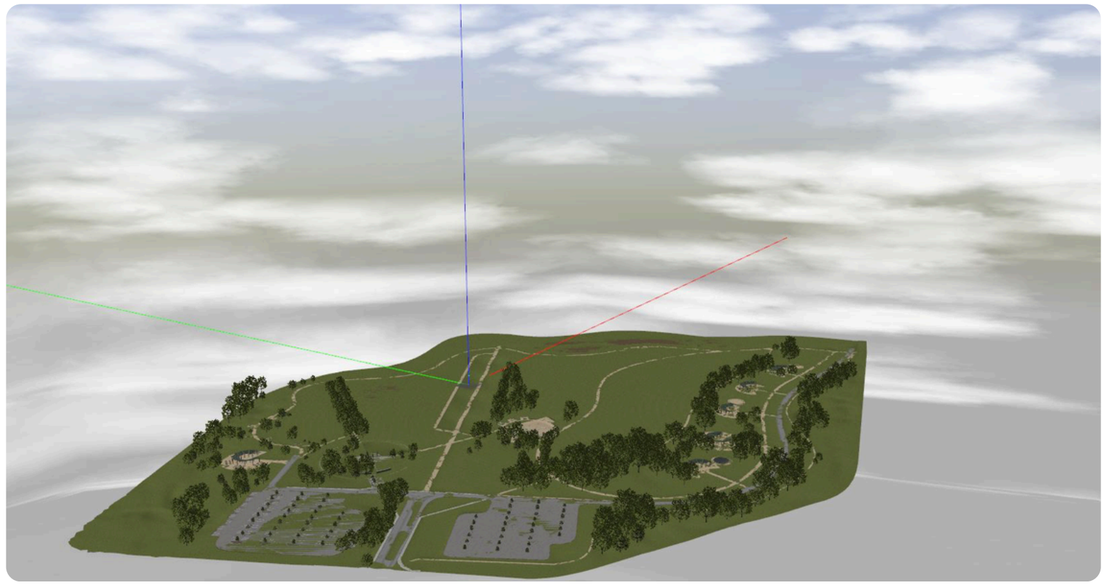
2D optical flow (RGB): classical Lucas-Kanade approach Simulation on ROS2

ROS2 Gazebo setup → Create a Gazebo environment with a flying environment → Image acquisition with OpenCV → Optical flow algorithm implementation and performance testing



Where we're at right now

2D optical flow (RGB)

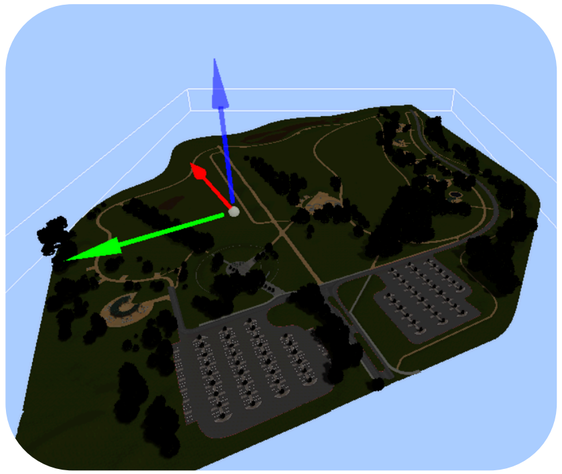


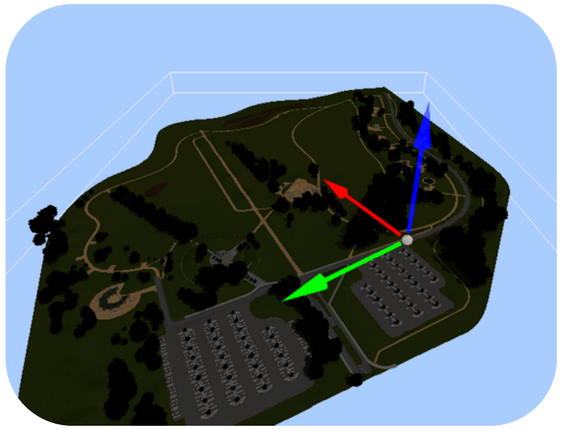
Environment : baylands_vegetation_right



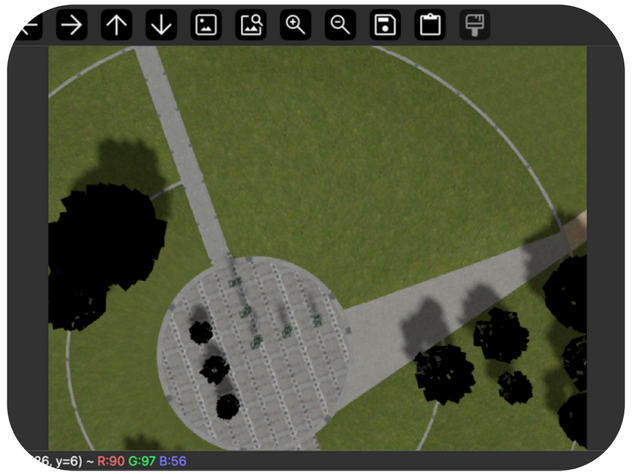
Drone : px4

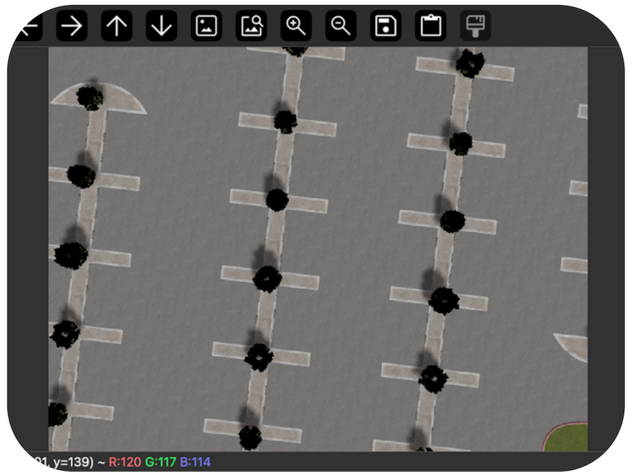
2D optical flow (RGB)



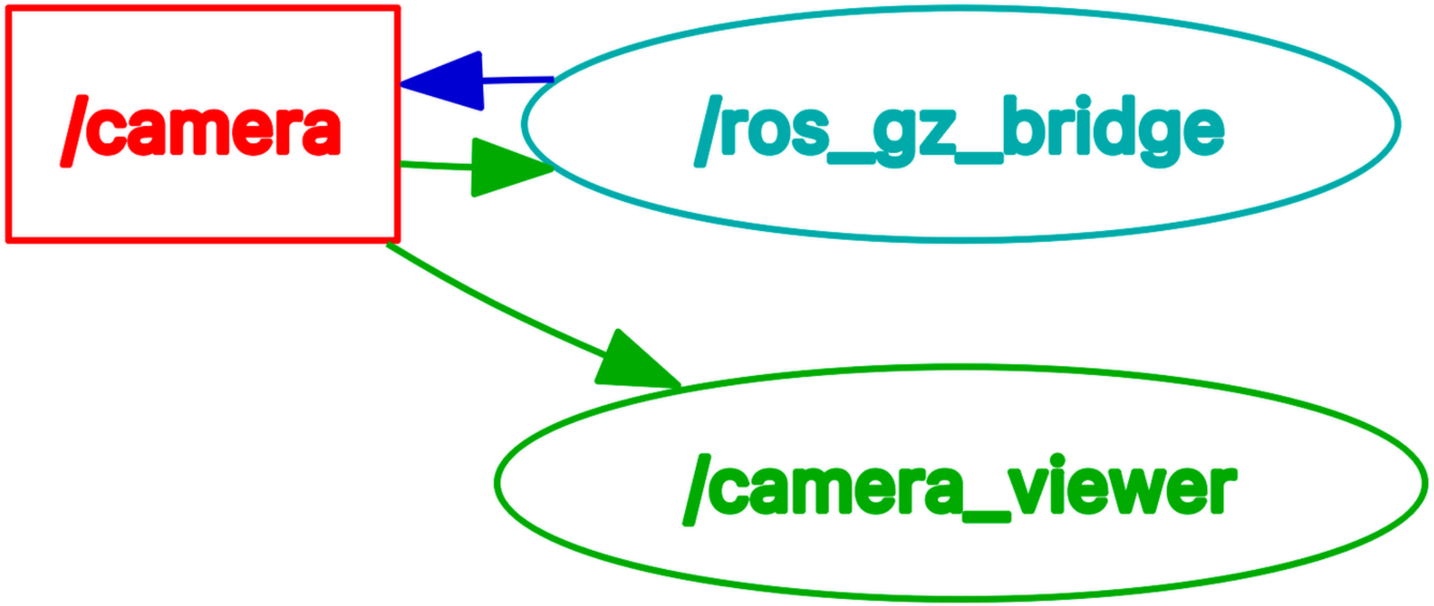


RGB Camera





2D optical flow (RGB)



Project roadmap - Current status

3D visual odometry (RGB-D): extension with depth integration

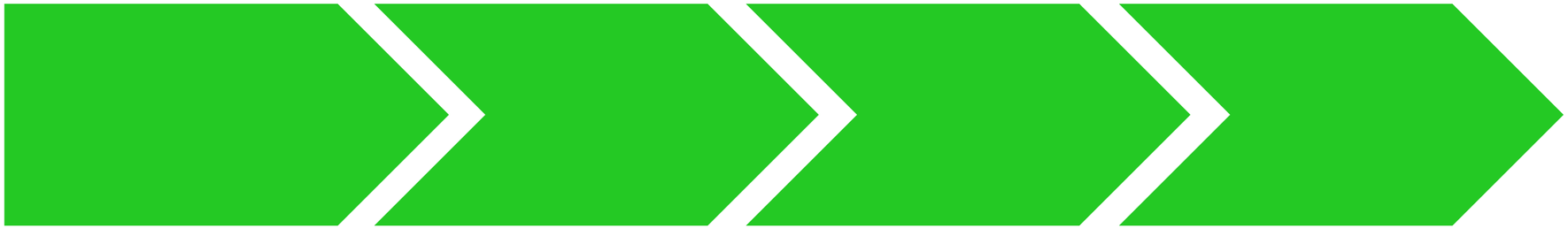




Where we're at right now

Project roadmap - Current status

Feature tracking via clustering: clustering-based approach to track features

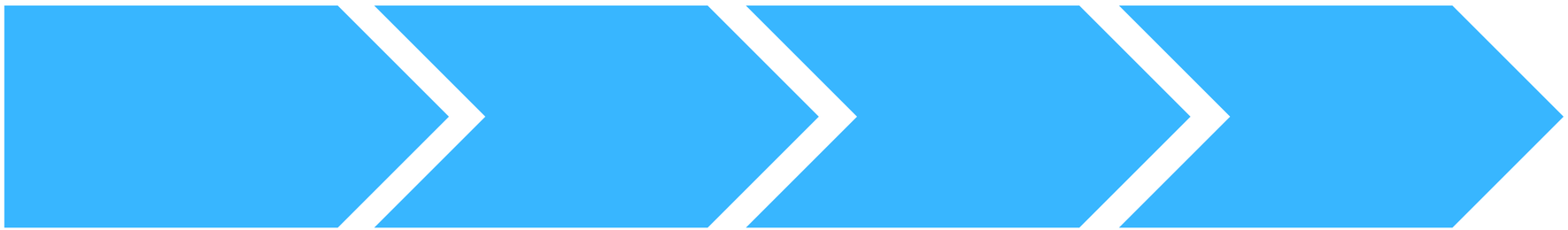




Where we're at right now

Project roadmap - Current status

SLAM: to be confirmed depending on progress



Where we're at right now

Technical environment





- **Simulation stack**: ROS 2 Jazzy + Gazebo Harmonic, on Ubuntu 24.04
- Drone model with a downward-facing camera, using the **PX4–Gazebo integration**
- **OptiTrack** motion capture system used as ground-truth reference
- **Git-organized workspace**: ros2_ws (src / build / install / log) alongside Data, Rapport, Presentation, Video folders



Progress so far

- Framing phase complete: ROS 2 Jazzy + Gazebo Harmonic + PX4 stack confirmed, Git repo structure initialized (ros2_ws, Data, Rapport, Presentation, Video)

- Camera simulation pipeline fully validated end-to-end: Gazebo → ROS 2 bridge → RViz2, image topic confirmed (640×480, rgb8)

Next steps

- Write a program to enable the drone to move autonomously within the simulation environment

- Use the Lucas-Kanade algorithm

Any questions?

